

Artificial Intelligence

AI2002

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GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Rules Representation
- Rules are often represented by bit strings (because they can be easily manipulated by genetic operators), but other numerical and symbolic representations are also possible
- Set of if-then rules:
 - Specific sub-strings are allocated for encoding each rule pre-condition and post-condition
 - Example: Suppose we have an attribute “Outlook” which can take on values: Sunny, Overcast or Rain

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Rules Representation
- We can represent it with 3 bits:
 - 100 would mean the value Sunny,
 - 010 would mean Overcast &
 - 001 would mean Rain
 - 110 would mean Sunny or Overcast
 - 111 would mean that we don't care about its value
- The pre-conditions & post-conditions of a rule are encoding by concatenating the individual representation of attributes

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Rules Representation
- Example:
 - If (Outlook = Overcast or Rain) and Wind = strong
 - then PlayTennis = No
 - can be encoded as 0111001
- Another rule
 - If Wind = Strong
 - then PlayTennis = Yes
 - can be encoded as 1111010

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Rules Representation
- An hypothesis comprising of both of these rules can be encoded as a chromosome 01110011111010
- Note that even if an attribute does not appear in a rule, we reserve its place in the chromosome, so that we can have fixed length chromosomes
- Note that 01110011111011 would mean, we do not care whether we play tennis or not

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Variable size chromosomes
- Sometimes we need a variable size chromosome; e.g. to represent a set of rules
- Example:
 - Suppose we are representing a set of rules by a chromosome
 - If $a_1 = T$ and $a_2 = F$ then $c = T$
 - If $a_2 = T$ then $c = F$
- If there is only the 1st rule in the set, we have 10 01 1
- If the rule set has 1st & 2nd rules then we have 10 01 1 11 10 0

Activity: Chromosome Encoding for Simplified Time Table

- Courses = 16 (C1, C2,C16)
- Teachers = 8 (T1, T2, T3,.....T8)
- Days = 5 (Mon, Tues, Wed, Thurs, Fri)
- Time-Slots = 5 (9-10, 10-11, 11-12, 12-1, 1-2)
- Classrooms = 4 (R1, R2, R3, R4)

- Chromosome encoding (Binary encoding)?
- Size of chromosome?

GENETIC ALGORITHM - Population Size

- Number of individuals present and competing in an iteration (generation)
- If the population size is too large, the processing time is high and the GA tends to take longer to converge upon a solution (because less fit members have to be selected to make up the required population)
- If the population size is too small, the GA is in danger of premature convergence upon a sub-optimal solution (all chromosomes will soon have identical traits). This is primarily because there may not be enough diversity in the population to allow the GA to escape local optima

GENETIC ALGORITHM - Fitness Function

- Also called “evaluation function”
- It is used to determine the fitness of a chromosome
- Creating a good fitness function is one of the challenging tasks of using GA

GENETIC ALGORITHM - Fitness Function

Example: Cookies Problem

Two parameters sugar and flour (in kgs). The range for both is 0 to 9 kgs. Therefore a chromosome will comprise of two genes called sugar and flour

5	1
2	4

The fitness function for a chromosome is the taste of the resulting cookies; values can be from 1 to 100

GENETIC ALGORITHM - Fitness Function

Example: Model Learning

Use GA to learn the concept Yes Reaction from the data

The fitness function can be **the score of the classification accuracy of the rule-set over a set of provided training examples** (number of training samples correctly classified by a chromosome)

Often other criteria may be included as well, such as the complexity of the rules or the size of the rule set

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

GENETIC ALGORITHM - Genetic Operators

- Genetic operators are applied to chromosomes that are selected to be parents, to create offspring
- Basically of two types: Crossover and Mutation
- Crossover operator create offspring by recombining the chromosomes of selected parents
- Mutation is used to make small random changes to a chromosome in an effort to add diversity to the population

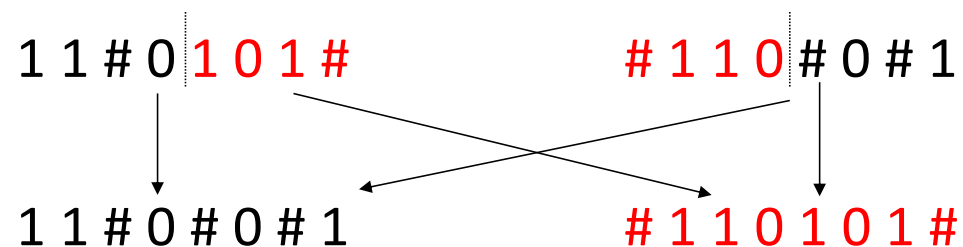
GENETIC ALGORITHM - Genetic Operators:

Crossover

- Crossover operation takes **two candidate solutions** and **divides** them, **swapping components** to **produce two new candidates**

GENETIC ALGORITHM - Genetic Operators: Crossover

- Figure illustrates crossover on bit string patterns of length 8
- The operator splits them and forms two children whose initial segment comes from one parent and whose tail comes from the other
- Input Bit Strings



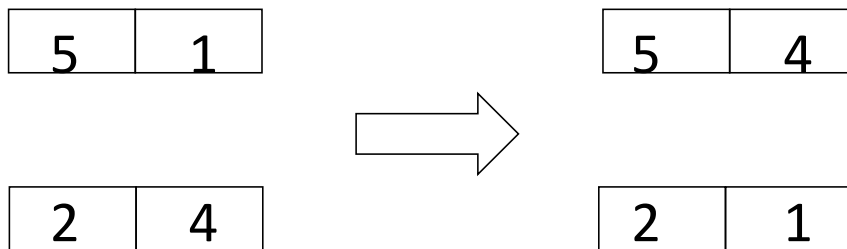
- Resulting Strings

GENETIC ALGORITHM - Genetic Operators:

Crossover

Two genes sugar and flour (in kgs)

Crossover operation on chromosomes



GENETIC ALGORITHM - Genetic Operators:

Crossover

- The place of split in the candidate solution is an **arbitrary choice**. This split may be at any point in the solution
- This splitting point may be **randomly chosen** or **changed systematically** during the solution process
- Crossover can unite an individual that is doing well in one dimension with another individual that is doing well in the other dimension

GENETIC ALGORITHM - Genetic Operators:

Crossover

- **Two types: Single point** crossover & **Uniform** crossover
- Single point crossover
- This operator takes **two parents** and **randomly selects a single point** between two genes to cut both chromosomes into two parts (this point is called **cut point**)
- The first part of the first parent is combined with the second part of the second parent to create the first child
- The first part of the second parent is combined with the second part of first parent to create the second child

100 00 10	1000001
111 0001	11100 10

GENETIC ALGORITHM - Genetic Operators: Crossover

- Uniform crossover
- The **value of each gene of an offspring's chromosome is randomly taken from either parent**
- This is equivalent to **multiple point crossover**

1000010

1110001

1010010

GENETIC ALGORITHM - Genetic Operators: Crossover (Variable size chromosomes)

Suppose crossover points (1, 8) happen to be chosen for the 1st parent
and crossover points (1, 3) happen to be chosen for the 2nd parent

1st parent 10 01 1 11 10 0

2nd parent 01 11 0 10 01 0

The resulting two offspring would be

11 10 0

and

00 01 1 11 11 0 10 01 0